Calculate the instantaneous power and average power absorbed by the passive linear network of Fig. 11.1 if

$v(t) = 330\cos(10t + 20°)$ V and $i(t) = 33\sin(10t + 60°)$ A

Answer: $3.5 + 5.445\cos(20t - 10°)$ kW, 3.5 kW.

Practice Problem 11.1

$p(t) = v(t)i(t)$

$P = \frac{1}{T}\int_0^T p(t)\,dt$

瞬时功率和平均功率的定义

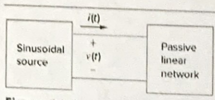

Figure 11.1
Sinusoidal source and passive linear circuit.

瞬时功率 $p(t) = v(t)\cdot i(t)$

$= 330\cos(10t+20°)\cdot 33\sin(10t+60°)$

$= \left[\frac{1}{2}\sin(20t+80°) - \frac{1}{2}\sin(-40°)\right]\cdot 330\cdot 33$

$= 3500 + 5445\cos(20t-10°)$ W

平均功率为瞬时功率的常数项：3500 W

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Practice Problem 11.5

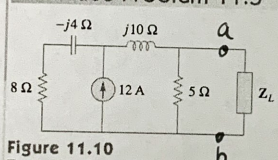

Figure 11.10
For Practice Prob. 11.5.

For the circuit shown in Fig. 11.10, find the load impedance $\mathbf{Z}_L$ that absorbs the maximum average power. Calculate that maximum average power.

Answer: $3.415 - j0.7317\ \Omega$, 51.47 W.

共轭匹配

对 a-b 左边电路进行戴维南等效：

① 开路电压

$$V_{Th} = 12\cdot\frac{(8-4j)}{(8-4j)+(5+10j)}\cdot 5 = 23.415 - 29.268j$$

$$= 37.482\angle -51.34°$$

② 等效阻抗：

$$Z_{Th} = (8+6j)\parallel 5 = 3.415 + 0.732j$$

$$\therefore Z_L = Z_{Th}^* = 3.415 - 0.732j$$

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$$P_{max} = \frac{|V_{Th}|^2}{8R_{Th}} = \frac{37.482^2}{8\times 3.415} = 51.424\ W$$

Practice Problem 11.11

从复功率导出其他所有功率量

For a load, $\mathbf{V}_{rms} = 110\underline{/85^\circ}$ V, $\mathbf{I}_{rms} = 3\underline{/15^\circ}$ A. Determine: (a) the complex and apparent powers, (b) the real and reactive powers, and (c) the power factor and the load impedance.

Answer: (a) $330\underline{/70^\circ}$ VA, 330 VA (b) 112.87 W, 310.1 VAR, (c) 0.342 lagging, $(12.541 + j34.46)\ \Omega$.

(a) 复功率 $S = V_{rms} \cdot I_{rms}^* = 330\angle 70^\circ = 112.867 + 310.099j$ VA

视在功率为复功率的模：330 VA

(b) 有功功率(平均功率)为复功率的实部：112.867 W

无功功率为复功率的虚部：310.099 VAR

(c) 功率因子 $pf = \frac{P}{S} = \frac{112.867}{330} = 0.342$

$Z = \frac{V_{rms}}{I_{rms}} = 12.541 + 34.455j$

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Practice Problem 11.15

$$C = \frac{Q_C}{\omega V_{rms}^2} = \frac{P(\tan\theta_1 - \tan\theta_2)}{\omega V_{rms}^2}$$

Find the value of parallel capacitance needed to correct a load of 140 kVAR at 0.85 lagging pf to unity pf. Assume that the load is supplied by a 220-V (rms), 60-Hz line.

功率因数校正

Answer: 7.673 mF.

$Q = 140K$ VAR

$C = \frac{Q}{\omega V_{rms}^2} = \frac{140K}{2\pi \cdot 60 \cdot 220^2} = 7.677$ mF

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